

Emotion-Driven Hypothesis for the Origins of Language

From Cross-Species Research to Neuroscience: Exploring the Evolutionary Pathways of Language

Theoretical Prelude

“All entities that transmit information, from mycelium to ants to birds and beasts, possess a form of language. Only when the volume of information reaches a critical threshold and is codified does it become what we, as humans, commonly define as 'language.' I argue that the true origin of language lies in the complexity of emotional modeling. From the perspective of human facial features and physiological structure—even in the prehistoric era before the advent of verbal language—humans exhibited a far greater complexity of emotional expression through facial muscles and bodily movements than any other species, including the unique proportion of the human sclera. When we look at all the intelligent species known today—cetaceans, felines, canines, elephants, primates—they each possess their own communicative systems. Yet, their differences in intelligence closely mirror the diversity of their emotional repertoires. I propose that it is precisely this strong correlation that led humans, under the evolutionary pressure to express increasingly nuanced emotions, to develop what we now call language.”

Note to readers:

As a non-native English speaker and independent researcher from Taiwan, I sincerely welcome any feedback, corrections, or suggestions.

If you find the hypothesis presented in this paper of interest and would be willing to assist me in submitting it to arXiv, my endorsement code is **CLXFOS**.

For any further inquiries, interest, or discussion, please feel free to contact me at:

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Your time, insight, and generosity are deeply appreciated.:

— Chun-Hao Chan

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Introduction | The Origin of Language: Not Merely Symbols, but the Outlet for Emotion

The origin of human language has long been a central topic at the intersection of linguistics, evolutionary biology, and anthropology. Existing mainstream theories tend to focus on the need for tool use (Stout & Chaminade, 2007; Arbib, 2005), the pressures of social cooperation (Dunbar, 1996; Tomasello, 2008), or the cognitive advancements brought by increased brain capacity. While these theories identify important contributing factors, they fall short of explaining deeper language functions—those involving emotional expression, narrative structure, and cultural transmission.

Moreover, these deeper functions often transcend the mere transmission of factual information. The capacity of language to express subjective emotional states, construct imagined realities, and transmit intergenerational narratives suggests that its evolutionary role is far more than utilitarian. These features imply that emotional pressure—not informational necessity—may have been the true catalyst for linguistic abstraction.

This paper proposes a new hypothesis for language evolution: the Emotion-Driven Language Evolution Hypothesis. This hypothesis asserts that human language did not emerge solely due to cognitive complexity, but was fundamentally catalyzed by the increasing nuance, stratification, and pressure of emotional expression in the human brain.

In early human societies, emotion functioned as a core mechanism for social bonding and survival. The accurate expression of internal emotional states was vital for group coordination, empathy, and conflict management. Language thus emerged primarily as a high-density symbolic outlet for emotional tension—far beyond a utilitarian system for exchanging objective information.

Importantly, emotional complexity is not determined by neural architecture alone. It is also closely tied to species-specific traits such as social attachment, inhibition of instinctual behavior, and degrees of group dependency (Barrett, 2017; De Waal, 2006; Coren, 1994). For example, in canines, breeds differ significantly in how frequently, deeply, and subtly they express emotions—a difference that correlates with their respective learning capacities and responsiveness to human interaction (Coren, 1994).

As emotional modeling in humans grew more sophisticated, language evolved from a basic communication tool into a symbolic system capable of simulation, abstraction, and future projection. This evolution required language to carry not just one function but at least two layers of logic: (1) the representation of emotion, and (2) an intention-driven reasoning process embedded within situational constraints. This dual structure is referred to as the 'Emotion–Intention–Reasoning Chain.' It serves as a key driver for linguistic abstraction and recursive structures. (Frith, 2006; Fitch, 2022). It is precisely when a communicative goal becomes "near but unattainable" that this chain is activated—emotion fueling intent, intent demanding inference, and inference shaping symbolic structure. This defines language as a tension-resolution mechanism rooted in psychological simulation.

Individual exposure to narrative input (e.g., fictional stories, role-based interactions) strongly influences the development of both linguistic and emotional faculties. Studies have shown that frequent fiction readers score higher on Theory of Mind tasks, indicating that narrative experience enhances one's ability to simulate and predict the emotions and motives of others (Mar et al., 2006; Kidd et al., 2018). Functional MRI studies also reveal that narrative comprehension activates emotion-related brain regions (such as the amygdala), language centers (Broca's and Wernicke's areas), and social cognition regions (rTPJ) in synchrony. This suggests that narrative understanding is a compound cognitive process combining emotional modeling and linguistic reasoning (Frith, 2006).

Narrative and emotion reinforce each other, deepening the logic of language structure. This study seeks to synthesize behavioral, cognitive, and neurobiological evidence to build a testable theoretical framework for emotion-driven language evolution. Through this lens, we may reassess the origin of language not as a mechanical tool of representation but as a cultural scaffold shaped by emotional needs.

Chapter 1 | A Broad Definition of Language and the Evolutionary Background of Communicative Needs.

1.1 Operational Definition of Language: Information Density, Symbolism, and Recursive Structure

The definition of language varies widely across disciplines. In linguistics, it is typically described as a system encompassing phonology, syntax, and semantics. In cognitive science, it is understood as a symbolic framework for representing and processing information. This study adopts a functional perspective: language is a high-density communicative system marked by symbolic substitution, syntactic arrangement, recursive layers, and contextual adaptability.

1.2 Information Transmission vs. Emotional Communication: The Convergence of Evolutionary Axes

Traditional theories of language evolution often emphasize the transfer of factual data—hunting signals, threat alerts, or spatial navigation. However, such accounts understate a central aspect of daily language use: emotional communication. While no study has definitively stated that over 70% of human speech is emotionally charged, numerous findings support the idea that emotional content dominates verbal interaction. Rimé (2009) observed that 88–96% of emotional experiences are shared through speech, usually within the same day. Emotional tone and lexical selection also influence conversational flow and semantic interpretation (Skerry & Saxe, 2015).

1.3 Pre-Language Systems and Evolutionary Continuity: Theoretical Foundations for This Hypothesis

Communication systems like fungal chemical signaling, ant pheromones, and bee dances all serve information transmission functions and can be considered broad precursors to language. However, these systems operate on simple stimulus-response mechanisms and lack the key properties of true language: the ability to combine elements, refer to abstract or displaced entities, and generate novel content beyond immediate context. For a true language system to emerge, more advanced internal modeling and recursive processing capabilities were necessary. This occurs particularly when the goal of an emotional intention is “nearly attainable yet out of reach,” triggering the inferential mechanism that gives rise to linguistic recursion.

This study proposes the "Emotion–Intention–Reasoning Chain" as the core scaffold of language origin. We argue that emotion serves as the primal driver of linguistic formation, while intention creates the adaptive pressure that necessitates linguistic recursion. It is only when emotional desire reaches a level of strategic ambiguity—where one must express a goal that is "near but unreachable"—that language enters the domain of reasoning. In this way, language emerges as a system for externalizing internally simulated logic based on emotional tension.

1.4 From the Tool Hypothesis to the Social Bonding Hypothesis: Integration and Limitations

The "Tool Hypothesis" posits that language evolved in conjunction with manual coordination and tool use, with overlapping neural substrates in Broca's area (Arbib, 2005; Stout & Chaminade, 2007; Greenfield, 1991). The "Social Bonding Hypothesis" views language as a more efficient substitute for grooming, helping maintain group cohesion through verbal exchange (Dunbar, 1996).

Yet, both theories fall short of explaining the dual-layered architecture of modern human language: its ability to express abstract logic and encode intricate emotional subtleties. This chapter concludes that while external ecological demands influenced language evolution, it was the internal emotional tensions—layered, unresolved, and socially consequential—that catalyzed the emergence of symbolic structure and recursive mechanisms in language.

The next chapter explores how emotional architecture itself imposed evolutionary pressure, making language not just possible, but necessary.

Chapter 2 | Emotional Complexity as a Potential Key Variable in Language Evolution

2.1 Definition and Core Components of the Emotional Model

Emotion is an internal need manifesting outwardly as behavior or conscious experience. Setting aside its acoustic or symbolic representations, even the fundamental requirements for sunlight, air, or water may be viewed as forms of emotion—what is commonly referred to as a sense of need, or, in more intense cases, craving.

Barrett (2017) suggests that emotion is essentially an “internal model of needs and environmental assessment,” with its expression varying by culture, species, and neural mechanisms. This diversity helps explain why emotion may serve as the seed of a symbol-based communicative system.

The term “emotional model” here refers to an individual’s capacity to perceive, categorize, label, and regulate both their own and others’ emotional states. This model encompasses three core abilities: (1) emotion recognition and differentiation; (2) emotion regulation and expression; and (3) the prediction and empathic understanding of emotions in social contexts. The advancement of these capacities generates increasing pressure for expressive communication, eventually driving the need for language.

Panksepp (1998) identified seven core emotional systems (such as SEEKING, FEAR, CARE, etc.), providing a clear neurobiological framework for emotion. This also indirectly supports the proposition that the complexity of emotional modeling could constitute a selection pressure for language evolution.

2.2 The Link Between Emotional Recognition Pressure and Language Development

As emotional models become more complex, primitive systems of facial expression and vocal tone are no longer sufficient to convey the subtleties of internal affective states. For instance, both “shame” and “regret” are negative emotions, yet the former is more concerned with self-worth, while the latter involves cognitive reflection on the impact of one’s actions on others. Without appropriate linguistic structures, such nuanced differences are difficult to communicate in social contexts, often resulting in misunderstanding and impaired group cognition.

Furthermore, behaviors such as jealousy, envy, and competition for affection—commonly observed in animals—serve as clear indicators of more complex emotional phenomena. While these behaviors exist across various species, the frequency, flexibility, and nuance with which they are expressed tend to correlate positively with the species’ level of cognitive plasticity and social intelligence. This reinforces the hypothesis that emotional complexity is not only a psychological trait, but also a driver of communicative sophistication.

De Waal’s (2006) observations confirm that chimpanzees and other primates exhibit pronounced social emotions, often intricately tied to social strategies and hierarchical dynamics. This provides indirect evidence that emotional modeling and cognitive development are mutually reinforcing, and potentially co-evolving.

Developmental psychology also supports this view. Emotional categorization and differentiation are not fully innate; rather, they require linguistic and contextual scaffolding to take shape. Hoemann et al. (2020) demonstrated that children rely on verbal labels and cultural input to construct clear emotional concepts from otherwise ambiguous sensory and bodily experiences. This provides compelling support for the argument that “language precedes emotional stabilization” and functions as an essential tool for constructing emotional representations and enabling emotional reasoning.

2.3 Biological and Evolutionary Mechanisms: The Evolutionary Significance of Human Sclera and Facial Musculature

Humans are the only primates whose exposed sclera (the white part of the eye) is conspicuously more visible than in other primate species. This trait markedly enhances gaze-following and joint attention, facilitating group-level emotional coordination and cooperative behaviors. Additionally, the evolution of a diverse array of facial muscles—including the frontalis, zygomaticus, and orbicularis oris—enables humans to express fine-grained differences and dynamic states in emotion.

Fridlund (1994) noted that the evolution of facial expressions serves not merely to release internal states, but more importantly, to transmit social signals and influence the behavior of others. This aligns with the current study’s proposition that “the emotional model serves as a source of evolutionary pressure for language.”

Barrett (2017) further emphasizes that emotion is fundamentally a predictive, context-dependent process constructed within social environments, rather than a set of fixed biological reflexes. This perspective strengthens the plausibility of language as an external structure for the emotional model.

Therefore, I propose that research into the origins of language should incorporate new observational frameworks such as “cross-species indices of emotional representation,” “emotional categorization precision,” and “emotion–intention–semantics linkage models.” These could provide novel statistical and comparative avenues for research on language evolution.

Chapter 3 | Cross-Species Comparison and the Proposal of an Empirical Framework

3.1 The Russian Fox Domestication Study and the Plasticity of Emotional Models

In a classic experiment spanning more than sixty years, Russian scientist Dmitry Belyaev led a domestication project in which wild silver foxes were selectively bred for tameness, ultimately producing a strain with dog-like social behaviors. The study not only observed behavioral changes (such as hand-licking, tail-wagging, and voluntary approach), but also found significant alterations in modes of emotional expression, facial muscle usage, and vocalization frequency range.

It is noteworthy, however, that while this experiment generated abundant behavioral and genetic data, no prior research has quantified or systematically compared “emotional density” itself across samples. Nonetheless, it is intuitively evident that domesticated foxes display a greater diversity of emotional behaviors, comprehend more human instructions, and exhibit heightened social interaction and problem-solving abilities—outcomes that can be regarded as “cognitive enhancement under a highly developed emotional model” (Belyaev & Trut, 1975–2019).

The findings indicate that emotional models are not fixed biological instincts but exhibit high plasticity. When domestication or attachment demands enhance the emotional model, the pressure for precision in communication increases correspondingly, leading to subtle adjustments in vocal and non-verbal expressive behaviors. Extrapolating from this mechanism, one can hypothesize that in highly socialized environments, individuals with more advanced “emotional representation, sharing, and simulation” abilities are more likely to develop communication systems with symbolic and grammatical features.

3.1-2 Dog–Cat Comparison: Case Studies in Animal Emotional Models and Linguistic Potential

Take, for example, dog breeds: intelligent breeds such as border collies and poodles clearly demonstrate emotional nuances (such as jealousy, seeking affection, or confusion), while less intelligent breeds tend to respond with a narrower behavioral repertoire (Coren, 1994). Although cats can express emotion, they generally rely on instinctual and defensive mechanisms and seldom exhibit a controllable emotional model. These comparisons show that differences in emotional modeling are closely correlated with cognitive plasticity.

The conclusion: The emergence of language is not only proportional to the complexity of information, but also directly linked to the differentiability, expressibility, and shareability of emotional structures. The pressure resulting from emotional model complexity may be the key driver in the transformation of primitive signaling systems into symbolic, structured language.

It is worth noting that this hypothesis does not conflict with existing research frameworks; rather, it builds upon theories of language evolution, affective neuroscience, and social cognition. Its core contribution lies in shifting the observational focus from cognition to emotion, and reframing the statistical approach to accommodate emotional modeling as a measurable evolutionary variable. Whereas most prior work has treated language primarily as a cognitive product, few studies have quantitatively operationalized or cross-species tested the role of “emotional complexity $\times$ expressive pressure”—a critical factor that remains significantly underexplored in comparative research.

Beyond the aforementioned animal studies, research on great apes' social behaviors also provides sporadic evidence for the positive correlation between emotional modeling capacity and linguistic potential. The reason such evidence has not entered mainstream discussions of language origins is not due to a lack of testability, but rather the absence of targeted, integrative research frameworks.

These comparisons highlight a fundamental insight: the degree of emotional modeling is strongly correlated with cognitive plasticity. Thus, the true driver behind the transition from primitive signaling systems to symbolic and grammatical language may lie in the increasing pressure imposed by complex emotional structures.

3.2 Designing Cross-Species Indices of Emotional Complexity and Their Applications

To operationalize the hypothesis of “emotional complexity $\times$ language evolution pressure,” I propose three cross-species observable indicators:

1. Emotional Expression Density Index: The number and frequency of distinct emotional expressions exhibited by an individual per unit time.
2. Intention Simulation Index: The capacity of an individual to predict another's behavioral tendency based on the other's emotional state.
3. Acoustic Modulation Variability Index: The range of modifiable features (such as pitch, rhythm, and loudness) in non-verbal vocalizations.

These, combined with established intelligence assessment methods and socialization indices, can be applied in comparative studies of chimpanzees, bonobos, dolphins, and canids. Preliminary inferences suggest that individuals displaying more frequent, varied, and socially engaged emotional behaviors tend to possess more complex communication systems—whether these be calls, gestural signals, or symbolic vocalizations.

3.3 Emotion-Driven Scenario Tasks as Triggers for Inferential Pressure

Building on the previous chapter's hypotheses, empirical studies may employ emotionally contextualized interaction scenarios to simulate language demands under various emotional combinations. For example, participants might be placed in scenarios involving misunderstanding, the need to explain, or unresolved affect, and then prompted to compensate or

persuade through language. The complexity of their output could be quantified by metrics such as syntactic recursion, lexical variation, and subjective semantic richness.

This methodology can be extended to populations with atypical development, such as individuals with developmental delays, autism spectrum disorder, or Asperger's syndrome, compared to typical controls. By examining language production under emotional-inferential pressure, one can test whether greater emotional understanding is associated with increased linguistic complexity. Similarly, educational interventions that integrate emotional and linguistic training can be evaluated for their impact on language and cognitive development.

Summary of the Chapter:

The empirical framework outlined here is already feasible at a preliminary level. From cross-species comparisons to human pragmatic task design, this approach enables multi-axial testing and falsification of the “emotion-driven language evolution” hypothesis, paving the way for its transition from theory to operational logic. In doing so, the hypothesis moves beyond conceptual theorization toward a concrete operational framework with high empirical tractability.

Chapter 4 | Comparison of Existing Theories of Language Origins and the Positioning of the Current Hypothesis

4.1 Mainstream Theories of Language Origins and Their Limitations

In the study of language origins, three major categories of mainstream theories can be identified:

1. **Tool-Use Cooperation Hypotheses:** These emphasize that language arose from the coordination, planning, and sequencing required for tool use, highlighting the link between language and action sequencing. For instance, Stout & Chaminade (2007) observed significant overlap in neural activity between tool-making behavior and classical language areas (such as Broca's area), suggesting that language may have originated from the cognitive and neural structures underpinning tool manipulation. While this theory accounts for the motor-control foundations of language, it does not explain the origins of rich emotional expression and symbolic meaning.
2. **Social Bonding Hypotheses:** Dunbar (1996) proposed that human language evolved as a substitute for grooming, serving as a more efficient means for maintaining social cohesion. This theory successfully accounts for the social function of language, but tends to neglect the inferential and internal modeling demands of language.

3. Cognitive Niche Hypotheses: Pinker (2010) argued that language arose as part of humans' "cognitive niche"—the ability to describe, predict, and model the world through abstract representations. This view emphasizes language as a tool for cognitive computation but gives less attention to the emotional and interactional pressures driving language evolution.

In addition, a crucial yet often separately discussed thread concerns mirror neurons and imitation. Discovered in the 1990s by Rizzolatti and colleagues, the mirror neuron system led Arbib (2005) to suggest that language originated from the neural mechanisms supporting action imitation and intention modeling. While this theory elucidates the deep connection between language and motor imitation, it faces the same challenge: Why do only humans, rather than other species with imitation capacities, develop fully recursive and symbolic language systems?

4.2 Imitation and Mirror Neuron Theories: Compensatory Mechanisms and Necessary Conditions

4.2.1 The Intersection of Imitative Ability and Emotional Demand: The Cross-Pressure Point for Language Evolution

This study offers an augmentation: While the mirror neuron system provides the neural substrate for imitation and action prediction, only the intersection of "imitative ability" and "intense sociality $\times$ emotional modeling pressure" forms the true threshold for language evolution. This helps explain why certain species with strong imitation skills (e.g., crows, parrots) can learn and reproduce vocal patterns, yet do not develop language systems with semantic recursion or symbolic logic. In contrast, species such as cetaceans, primates, and canids—possessing both strong emotional attachments and hierarchical social structures—exhibit complex social-emotional interactions and communication behaviors. Thus, while imitation is widespread in the animal kingdom, only the superimposition of emotional sharing and intention simulation creates the evolutionary selection pressure necessary for language generation (Arbib, 2005; Rizzolatti & Caruana, 2023).

This perspective aligns with Tomasello's "shared intentionality" theory, which posits that the core evolutionary driver of language is the human capacity for cooperative action and intention inference—thus establishing a theoretical bridge between the present model and other leading frameworks (Tomasello, 2008).

4.2.2 Counterexamples and Supplements: Avian Vocal Systems

Birds (such as parrots, robins, and thrushes) display remarkable abilities for vocal imitation and modulation, including pitch, rhythm, mimicry, and intonation. However, these vocal behaviors typically serve limited purposes such as courtship, territory defense, or alarm signaling, and there is insufficient evidence that they are used for complex emotional expression or internal-state communication. Most avian vocalizations remain at the level of stimulus-driven response. Even parrots capable of mimicking human speech mainly engage in acoustic playback, rather than

generating recursive, semantically structured utterances. This observation supports the view that mirror neurons and intention simulation are necessary, but not sufficient, for the emergence of language. Pepperberg's famous subject Alex, a grey parrot, demonstrated limited semantic mapping and classification, but did not develop the capacity for flexible syntactic generation, nor the ability to linguistically mediate complex internal emotional contradictions—functions that define true symbolic language in humans. (Pepperberg, 2002).

4.3 The Complementary and Integrative Position of the Present Hypothesis

This study does not reject the foundational insights of existing theories, but rather proposes the “emotional origin of language” as a crucial supplement. Language is not merely an extension of tool use, nor simply a product of gossip and social grooming, but is the crystallization of an “emotion–intention–reasoning chain,” fulfilling the symbolic communication demands arising under intense emotional and inferential pressure. Deacon has discussed the symbolic nature of language, noting that language and the human brain co-evolved to enable semantic structure and abstraction. This view also underpins the current model's explanation of the neural and cultural sources of “symbolic communicative pressure” (Deacon, 1997).

Although the present hypothesis is primarily grounded in the role of emotional complexity as a source of pressure on language, it is worth noting that neuroevolutionary or epigenetic approaches may further reveal the imprint of this pressure at the physiological level (Jablonka & Lamb, 2005; see also Chapter 5).

4.4 Integrative Frameworks and Comparative Tables

4.4.1 [Figure 1: Integrative Framework of Language Origin Theories]

(1) Source of Evolutionary Drive → (2) Core Mechanistic Pathway → (3) Effects on Language Features

- Tool-use Cooperation: Hand – eye coordination → Internalization of action sequences → Verb systems and syntactic order
- Social Bonding: Group maintenance needs → Substitution of contact and intimacy exchange → Emotional vocabulary, address terms, connectors
- Cognitive Niche: Strategic thinking → Formation of symbolic systems → Abstract concepts, hypothetical mood
- Emotion-Driven Hypothesis: Emotional differentiation pressure → Emotion – intention – reasoning chain → Narrative simulation structure, subjective vocabulary

4.4.2 Comparative Table: Integrative Summary of Theories

Theory	Focus / Axis	Mechanism / Language Features	Verifiability	Emotional Scope	Narrative Scope	Main Limitation	Contribution of This Hypothesis
Tool-use Hypothesis	Motor-centric	Action coordination, syntax	★★★★☆	★★☆☆☆ ☆	★★☆☆☆ ☆	Lacks emotional motivation	Adds emotional/narrative pressure
Social Bonding Hypothesis	Social-centric	Group cohesion, vocabulary repetition	★★★★☆ ☆	★★★★☆ ☆	★★☆☆☆ ☆	Lacks inferential depth	Adds intention reasoning / internal tension
Cognitive Niche Hypothesis	Abstraction	Symbolic modeling, abstract reasoning	★★★★☆ ☆	★★☆☆☆ ☆	★★★★☆ ☆	Ignores emotional source	Integrates reasoning and emotion
Emotion-Driven Hypothesis	Emotion-centric	Differentiation, reasoning chain	★★★★★ ★	★★★★★ ★	★★★★★ ★	—	Highlights multi-layered pressure

Conclusion of the Chapter:

By incorporating emotional modeling as a central factor in language evolution, this hypothesis effectively fills gaps in the three major mainstream theories and enhances the integrative framework. By explicitly positioning emotional modeling as a core evolutionary pressure, this framework offers a critical corrective to existing paradigms—clarifying the missing link between emotional tension and symbolic expression in language development.

The next chapter will explore potential practical applications of this hypothesis in education, AI, and clinical intervention.

Chapter 5 | The Evolutionary Logic of Human Emotional Contradictions and the Pressure for Language Generation

5.1 Definition of Five Levels of Emotional Complexity (L1–L5)

Most existing theories on the origin of language have tended to treat "emotion" as a background factor supporting the social function of language. However, this study contends that emotion is not merely an accompaniment to language, but rather the primary driving force behind its emergence and evolution—especially as emotions enter stages of internal conflict and tension, thereby heightening the necessity for language as a tool for resolution.

Emotional complexity can be broken down into five distinct levels:

- **L1:** Physiological Needs Level – This refers to basic physiological needs and reactive responses, such as hunger, satiety, fear, and drowsiness.
- **L2:** Emotional Differentiation Level – The capacity to distinguish subtle emotional states such as happiness, comfort, regret, and shame.
- **L3:** Emotional Layering Capacity – The ability to simultaneously experience mixed emotions, such as love and hate, or dependence and fear.
- **L4:** Emotional Contradiction and Tension Management – The capacity to handle typical dilemmas involving internal conflict, such as “wanting but being unable to have,” “not wanting but having to do,” or making sacrifices among multiple options.
- **L5:** Memory Imprinting and Reconstruction Ability – The capacity to internalize these experiences as cultural memory and narrative structures within language.

These hierarchical levels serve as concrete sources of pressure for the construction of the “Emotion–Intention–Reasoning Chain,” and fundamentally distinguish the evolutionary pressures on language experienced by humans from those faced by other species.

Furthermore, according to Barrett’s “theory of constructed emotion,” emotion itself is a psychologically constructed category shaped dynamically by context and language, rather than a fixed physiological reflex. This perspective aligns with the present study’s core proposition that “language not only expresses emotion but is a necessary mechanism for the categorization of emotional experience.” It also supports the “Emotion–Intention–Reasoning Chain” as a foundational model for the pressures driving linguistic structure (Barrett et al., 2020).

The five-step model of emotional complexity proposed here is also highly consistent with the findings of Hoemann et al. (2020) on the co-development of language and emotional categories in children. Their research shows that without the intervention of language, individuals struggle to internalize and categorize complex emotional experiences, which in turn limits the emergence of advanced emotional logic and reasoning within language.

5.2 Emotional Contradictions as the Key Threshold for Language Evolution

This study proposes that the emergence of grammatical and symbolic language in humans—but not in other highly social species—is fundamentally rooted in the persistent need to model and resolve emotional contradictions over time. For instance:

- The invention of lying requires that language be able to distinguish between telling and withholding, truth and falsehood.
- Social phenomena such as indirectness, reluctance, or the act of “leaving things unsaid” show that language must accommodate the packaging and rhetorical expression of conflicting emotions.
- Dilemmas like “love versus responsibility” or “loyalty versus the pursuit of freedom” turn language into a vehicle for value hierarchies and personal transformation.

All these language use scenarios emerge because humans routinely experience psychological tension between behavioral choices and emotional feedback—a tension that can only be fully articulated and negotiated through language. This psychological “inner pressure” is thus one of the major driving forces behind language evolution. Such usage scenarios illustrate that language arose not merely for communication, but as a way to linguistically represent and resolve the internal tensions that emerge when behavioral choice and emotional feedback collide.

5.3 Why Do Only Humans Exhibit “Emotional Language”?

As noted in Chapter 3, other species with sophisticated mimicry or vocal abilities—such as parrots, cetaceans, and primates—may be able to replicate sound signals, but they do not display a need for language that handles internal emotional conflicts. Most vocal or behavioral imitation in these species serves simple social functions, such as courtship, signaling danger, or maintaining group contact, but lacks the multi-layered subjective intention and emotional tension found in human language.

Thus, the true evolutionary selection pressure for language may not have been simple information exchange, but rather the need to express and share subjective intention and internal emotional contradictions.

5.3-1 Self-Modeling and Social Mirroring as Underlying Drivers of Linguistic Pressure

Despite the high degree of sociality and intelligence in species such as elephants and non-human primates, none have evolved language systems with true grammar and symbolic logic. This study further argues that the critical factor is not only cognitive capacity or the complexity of social

behavior, but the emergence of a psychological need to construct the self in the mirror of others' minds.

In human societies, individuals routinely engage in metacognitive operations that reconstruct their own narrative and social position from the perspective of others. This “how am I perceived by others” process demands that language accommodate the sequencing of narratives, selective disclosure of emotional motives, and the dynamic adjustment of communication strategies according to the listener's identity. Such skills require not only intention simulation, but also the ability to reason about others' expectations and social hierarchies.

This resonates with Dawkins' (1976) concept of “strategic self-interest” and its cognitive projections in interpersonal language use. Language thus becomes not just a tool for cooperation, but a mechanism for individuals to shape their self-image within the mental models of others. As language becomes the stage for social competition and reconfiguration of identity, it evolves complex structures—such as symbol systems, syntax, prosody, and role-based pragmatics—to manage how messages are “received and understood” (Dawkins, 1976). This cognitive mechanism requires that language perform three key functions: controlling the order of narration, selectively disclosing emotional motives, and dynamically adjusting communicative strategies based on the listener's presumed perspective.

This perspective further reinforces the “reasoning” dimension of the Emotion–Intention–Reasoning Chain: reasoning is not only about predicting external events, but also about how the self is recognized in the minds of others—a key source of simulation pressure.

5.4 Language as an Outlet for Inner Tension

Language is not a cold, utilitarian tool, nor merely an externalized form of collaboration or knowledge. Rather, it serves as the outlet device for the complex internal contradictions of the human mind—it is the only system through which our conflicting selves can find expression in words. As emotional complexity rises, the symbolic, logical, and narrative aspects of language become evolutionary necessities.

Furthermore, the emergence of language as an expressive outlet has gradually reshaped the boundaries of how humans perceive and name their emotions. Emotion is not just the starting point of language, but also becomes a domain in which language itself is refined and stratified, resulting in a feedback loop. This dynamic suggests that language and emotion have likely entered a phase of mutually reinforcing co-evolution, rather than a one-way causal relationship (for further discussion, see subsequent chapters and the extended argument in Chapter 9).

5.5 Epigenetics: The Hypothesis of Intergenerational Emotional Imprints in Language Expression

In the field of epigenetics, studies of social insects such as ants and bees provide a critical perspective: differences in behavior and role expression are often not the result of changes in genetic sequence, but rather are activated by specific environmental pressures that trigger the expression of certain genetic segments (Jablonka & Lamb, 2005). This phenomenon suggests that phenotypic differences may originate from the long-term imprinting of experience, behavior, and environmental stimulation.

Recent advances in molecular neuroscience further elucidate the biological pathway from gene regulation to neural differentiation. For instance, Taniguchi et al. (2012) demonstrated that the *Nolz-1* (*Zfp503*) gene, by suppressing *Dlx1/2* expression, promotes the differentiation of neural progenitor cells in brain regions such as the striatum. Such findings provide concrete evidence linking gene regulation to the neurobiological basis of language and emotional capacities. For further molecular mechanisms, see Section 6.2-1.

Multiple studies in language development and neuroscience have shown that there exists a critical period for the acquisition of human language abilities. If a child does not receive sufficient linguistic input during early childhood (approximately ages 2–7), the declining plasticity of the brain's language areas will make it exceedingly difficult to acquire full linguistic competence—well-documented in cases of so-called “feral children.” This effect is observable not only at the level of neural network development but is also consistent with findings in epigenetics: environmental stimuli regulate gene expression and synaptic formation, and insufficient language exposure can suppress relevant gene switches, thereby affecting cross-generational development of language ability and the capacity to adapt to modern society.

Reviews by Hensch (2005) further indicate that the opening and closing of critical periods for brain development, such as in language learning, are closely linked to epigenetic mechanisms like DNA methylation. Environmental stimuli—such as language exposure and emotional interaction—can induce long-term structural changes in neural circuits during this period, resulting in “imprints” on gene expression patterns that influence cognitive and linguistic abilities across the lifespan. Such experience-driven molecular memories echo the transgenerational epigenetic mechanisms described by Jablonka & Lamb.

Extending this perspective to the origins of human language, this study proposes a future research direction: In early human groups, would long-term exposure to high-tension emotional states—such as suppressed expression, imprecise communication of emotion, or ambiguity leading to misunderstanding—produce inheritable epigenetic regulation at both the neural and genetic levels? This hypothesis invites a bold line of interdisciplinary inquiry: could chronic emotional tension—particularly the persistent inability to articulate intent—leave inheritable neuroepigenetic marks that scaffold future linguistic capacity?

Such an approach not only provides a biological mechanism to extend the “emotion-driven language evolution hypothesis,” but also offers a promising direction for future research that combines neural plasticity, developmental psychology, and epigenetics. If it can be further

demonstrated that “chronic emotional pressure × social demands” can regulate gene expression, then language should be regarded not merely as a cultural or cognitive product, but as a phenotypic tool of experiential inheritance.

Chapter 6 | Structural Models of Emotional Language Generation and Visualization

6.1 Mathematical Formalization of the Five-Level Emotional Complexity Model (L1–L5)

This chapter builds upon the framework presented in Chapter 5 by establishing a mathematically formalized, hierarchical structure for the dynamic factors driving language evolution. The aim is to construct an operational, visualizable, and quantifiable model of the key pressures underlying language emergence. The emotional complexity (“Ec”) axis is further refined here as a five-level model (L1–L5), serving as the main evolutionary driver for language:

- **L1:** Primary Physiological Needs – Basic instincts and immediate responses such as pain, hunger, or sexual drive, which are universally present across species.
- **L2:** Singular Emotional Responses – Basic social emotions such as happiness, anger, or fear, which are externally expressed but not oriented toward reasoning or abstraction.
- **L3:** Restricted and Delayed Emotions – Emotions that involve deferred gratification, anticipation, or behavioral continuation, as in “wanting food but being unable to obtain it,” which may drive tool use and problem-solving behaviors.
- **L4:** Value Conflict and Social Reasoning Emotions – Emotions such as shame, regret, empathy, or responsibility conflict, which require Theory of Mind and hierarchical modeling abilities.
- **L5:** Narrative and Symbolic Regulation – The evolution of language, imagery, and cultural symbol systems as mechanisms to process and mediate complex emotional tensions.

(The model begins at L1; L0 would represent a purely unconscious state and is therefore not included in cross-level comparisons.)

This formalized hierarchy not only explains the sources of emotional pressure driving human language generation but also delineates clear boundaries between species along the evolutionary spectrum of language potential. The model provides a robust foundation for both horizontal (cross-species) and vertical (evolutionary) comparative analysis.

6.2 The Triadic Interaction Model: $Ec \times Cn \times Ns$

The emergence of language is not driven solely by emotion; it also requires the integration of cognitive architecture and simulation abilities. To account for these interacting forces, we propose the following triadic interaction model:

- **Ec (Emotional Complexity):** Based on the L1 – L5 emotional pressure hierarchy, this dimension represents the capacity for emotional modeling and the internal representation of tension.
- **Cn (Cognitive Structure):** This includes abilities such as mirroring, causal reasoning, tool use, and symbolic abstraction—essential components of cognitive architecture supporting language.
- **Ns (Narrative Simulation Ability):** The ability to simulate scenarios beyond the present—such as the past, future, or hypothetical contexts—is the core of narrative and linguistic capacity.

These three axes form a mutually amplifying relationship. The potential for language generation (Lg) can be predicted by the following functional equation:

$$Lg = f(Ec \times Cn \times Ns)$$

Each parameter can be estimated through behavioral observation, cognitive testing, or neuroimaging results, and may be evaluated across species or clinical contexts for model validation.

6.2-1 Neural Correlates Supporting the $Ec \times Cn \times Ns$ Model

To further substantiate the evolutionary plausibility and biological grounding of the “ $Ec \times Cn \times Ns$ ” triadic model, this section maps each variable to corresponding neural structures, illustrating the strong alignment between the internal pressures for language generation and the foundations of neural plasticity.

A. Emotional Complexity (Ec) $\times$ Limbic System and Social Regulation Centers

The neural correlates of emotional complexity are concentrated in the limbic system, particularly the amygdala, anterior cingulate cortex (ACC), and insula. These regions are involved in primary emotional perception, pain representation, conflict detection, and the integration of internal emotional states. When emotions reach the stage of contradiction or dilemma (L4–L5), the ACC and insula play key roles in behavioral decision-making, moral judgment, and emotional labeling. Barrett’s work also emphasizes that emotion is a predictive, context-dependent construction that relies on memory and context, linking the hippocampus to semantic generation.

B. Cognitive Structure (Cn) × Mirror Neuron System and Executive Function Centers

The intersection of language evolution and cognitive structure is especially pronounced in the Broca's area (BA44/45) and the dorsolateral prefrontal cortex (DLPFC). Broca's area governs both speech planning and syntactic composition and has deep links with imitation, mirroring, and action-sequence simulation, aligning with Rizzolatti's "mirror neuron system" hypothesis (Arbib, 2005). The DLPFC is responsible for behavioral inhibition, causal prediction, goal planning, and conflict regulation—critical for transforming emotional contradiction into strategic choice.

The co-activation of these regions underpins the "tool reasoning × syntactic integration × strategic thinking" functionalities that are central to Cn, making them vital evolutionary nodes in language development.

C. Narrative Simulation (Ns) × Narrative and Theory of Mind Brain Regions

The neural substrate for narrative simulation involves the right temporoparietal junction (rTPJ), medial prefrontal cortex (mPFC), and posterior cingulate cortex (PCC). These areas form the core network for Theory of Mind, temporal sequencing, and contextual simulation, collectively described by Mar et al. (2009) as the "narrative comprehension system." These regions are consistently activated during story reading, character empathy, and role-based strategic judgments, closely linking narrative logic and linguistic strategy generation.

This system interconnects with the DLPFC, Broca's area, and ACC to create a cross-network of language, narrative, and emotion, highlighting that language generation is not purely computational but emerges from the integration of emotion, strategy, and simulation.

6.2-2 Three Key Challenges in the Evolution of Language Areas and the Integrative Explanation of the Ec × Cn × Ns Model

The Ec × Cn × Ns triadic interaction model constructed in this study not only serves as a framework for the evolutionary pressures underlying language generation, but also addresses three long-standing challenges in contemporary neurobiology and the neuroscience of language:

1. Why are Broca's and Wernicke's areas so highly developed in humans?

Traditional localization of language functions comes from studies of aphasia and neuroimaging: Broca's area is linked to speech production and syntactic composition, while Wernicke's area is involved in semantic decoding and lexical retrieval. However, even among other primates, comparable structures do not reach the scale or level of activation seen in humans (Fadiga et al., 2009; Binder et al., 2009).

Our model proposes that the remarkable development of Broca's area is tied to its role in "action simulation × sequential reasoning," corresponding to the Cn (cognitive structure) dimension. Wernicke's area, meanwhile, supports "contextual parsing × emotional interpretation," linking

the $E_c \times N_s$ axes. These dual regions are the product of the co-evolution of narrative construction and emotional reasoning.

2. Why do language areas and emotion-related areas (such as ACC, amygdala, insula) frequently co-activate?

Research shows that, during language production tasks—especially those involving emotional or social content—the anterior cingulate cortex and amygdala are frequently co-activated with language regions (Ochsner & Gross, 2005; Etkin et al., 2011).

This hypothesis offers a rational explanation: Language, in essence, is not merely about information output, but about processing “subjective emotional tension $\times$ predictive strategy simulation” in social contexts. The language system thus naturally fuses emotional perception with semantic generation.

3. Why can human language simulate non-present scenarios?

Conventional tool and imitation theories explain language’s advantages in real-time interaction (Arbib, 2005; Tomasello, 2008) but struggle to account for humans’ unique capacity to describe hypothetical events, imaginary characters, or alternative realities.

This ability is identified here as a result of “narrative simulation” (N_s), supported by regions such as the right temporoparietal junction (rTPJ), medial prefrontal cortex (mPFC), and posterior cingulate cortex (PCC)—all key hubs for Theory of Mind and narrative comprehension (Spreng et al., 2009).

Summary:

The $E_c \times C_n \times N_s$ triadic model provides a coherent framework integrating behavioral motivation, biological evolution, and cognitive construction in the origin of language. It also explains why the human language system is so closely intertwined with emotion and simulation: Language evolved as a narrative tool to manage “tension $\times$ prediction $\times$ role reconstruction” pressures. This integrative framework bridges evolutionary neuroscience with cognitive linguistics. It offers a viable pathway for advancing empirical studies on language origin.

6.3 Cross-Species Language Potential Comparison Table (Dual-Axis Example)

The following table uses the five-level emotional complexity model as the primary axis, supplemented by cognitive structure and simulation ability, to concretely predict language generation potential across species. This comparative framework can guide the design of behavioral experiments and the evaluation of language potential in AI models.

Species	Emotional Complexity (Ec)	Cognitive Structure (Cn)	Narrative Simulation (Ns)	Language Generation Propensity (Lg)
Humans	L1–L5	High	High	Very High
Chimpanzees	L1–L3	Medium	Low	Medium
Cetaceans	L1–L3	Medium	Medium	Medium
Grey Parrots	L1–L3	Medium	Very Low	Low
Wild Dogs	L1–L2	Low	None	Very Low
Domesticated Dogs	L1–L3	Medium	Low	Low

This table is anchored on the five-tier emotional complexity model and augmented by cognitive and simulation capacities. It concretely displays the range of predicted language generation potential and can be used as a reference for designing behavioral research and evaluating the potential of AI language models.

6.4 Applications of the Model and Directions for Empirical Research

This model, which integrates the axes of emotional pressure, cognitive conditions, and simulation ability, demonstrates significant potential for both research and applied domains:

- **Language Neuroscience:** Utilizing fMRI to observe co-activation patterns between brain regions associated with Ec (emotional complexity) and Ns (narrative simulation).
- **Animal Cognitive and Behavioral Science:** Monitoring changes in Ec – Lg (emotional complexity to language generation potential) to assess the effects of training and socialization across species.
- **AI Language Generation Systems:** Designing artificial language engines that incorporate emotional complexity and narrative simulation capabilities. For example, implementing hierarchical syntax for scenario-based emotional priorities, enabling machines to encode multi-level emotional and narrative tension similar to the human brain, thereby achieving dynamic reasoning and contextual adaptation.
- **Language Education and Special Education:** Using an individual's Ec level to diagnose language disorders and weaknesses in narrative logic, and developing targeted intervention programs for optimal development.

6.5 Summary: From Tension to Language—A Logical Closed Loop

The evolution of language is no longer understood as a mere byproduct of information exchange, but rather as the inevitable outlet for the accumulation of internal emotional pressure and the demand for reasoning. The framework constructed in this chapter—comprising the five-level model of emotional complexity, the triadic interaction model, the predictive equation, and the cross-species comparison table—offers a closed-loop structure with both theoretical depth and experimental utility. It opens multiple avenues for research on the origins of language, connecting neural plasticity, narrative simulation, and artificial intelligence.

Furthermore, the triadic interactive model ($E_c \times C_n \times N_s$) may, over long evolutionary timescales, give rise to a positive feedback loop of mutual reinforcement (for further discussion, see the extended argument in Chapter 9).

Chapter 7 | The Cultural Co-construction Model of Narrative Texts and Linguistic Ability

In the developmental trajectory of human cognition and language, emotionally charged narrative situations have not only served as critical stimuli for the formation of linguistic abilities but have also played a profound role in shaping higher-order intelligence capacities such as Theory of Mind, empathy, and role-based simulation. This chapter synthesizes evidence from behavioral science, neurodevelopment, and language education to demonstrate how the model of “complex emotions $\times$ contextual narrative input” acts as a developmental bridge between language evolution and cognitive growth. Furthermore, it argues that narrative-based language plays a pivotal role in enhancing social cognition and creativity at the individual level.

7.1 How Narrative Texts Stimulate the Development of Language in Early Childhood

The term contextualized narrative input refers to language materials that contain clear situational contexts, role interactions, emotional tension, and event progression—in other words, narrative linguistic content grounded in concrete scenarios. Recent research in language development and cognitive science has confirmed that such inputs significantly enhance children’s language comprehension, social cognition, and inferential abilities (Mar et al., 2006; Snow, 1983; Barsalou, 2008; Rowe & Weisleder, 2020; Kidd & Castano, 2013).

Building upon this, the present study proposes a corollary hypothesis: the complexity of language is not solely constrained by phonological or syntactic competence, but is fundamentally shaped by an individual’s ability to comprehend and reconstruct narrative tension. When children are exposed to a high volume of narrative texts during their critical periods of language acquisition,

they exhibit significantly accelerated development in domains such as cognitive prediction, logical reasoning, and emotional processing. This is not merely a process of linguistic acquisition but rather an early-stage activation of narrative simulation capacity.

Empirical findings suggest that around the age of 4–5, children begin developing basic Theory of Mind—the ability to construct models of others’ mental states. This capacity is foundational to understanding stories and empathizing with fictional characters. Exposure to narratives rich in emotional complexity and character-driven conflict (such as illustrated books or children’s stories) during this period has been shown to significantly increase activation in key brain regions, including the prefrontal cortex, the anterior cingulate cortex (ACC), and the temporoparietal junction (rTPJ) (Mar et al., 2006).

7.2 Empathy Development, Narrative Simulation, and the Neural Interaction Model of Language

Studies show close neural connections between brain regions involved in language and reasoning—particularly the dorsolateral prefrontal cortex (DLPFC)—and regions responsible for emotional inference, such as the insula and amygdala. The demands of contextual storytelling provide an ideal neural stimulus, enabling children not only to acquire vocabulary but also to engage in simulated perspective-taking and emotional conflict resolution (Howard, 2006; Barsalou, 2008).

Research in neurolinguistics and neuropsychology suggests that emotional intensity modulates memory consolidation via the amygdala, enhancing the retention of emotionally charged words, sounds, and expressions in both individual and cultural memory (McGaugh, 2004; Pessoa, 2008). This phenomenon helps explain the enduring cultural impact of emotionally intense utterances—such as exclamations, spells, group slogans, and chants—and their role in the symbolic crystallization of language. It also sheds light on how emotional memory enhances cross-generational transmission of vocabulary.

Furthermore, sustained engagement with narrative reasoning and emotional simulation has been shown to increase cortical thickness and functional connectivity in relevant brain regions (Ochsner & Gross, 2005; Etkin et al., 2011). These findings point to the existence of a “neuroplastic window” during childhood and suggest that early language education should prioritize emotionally engaging and cognitively challenging input to optimize linguistic and social development.

This chapter hypothesizes that narrative simulation mechanisms are not merely cultural byproducts, but fundamental cognitive drivers that shaped the recursive and symbolic structures of human language.

7.3 The Role of Fiction in Structuring Human Lifespan Narratives Across Civilizations

Another key factor lies in the extended lifespan of humans. Compared to other primates, humans exhibit a notably longer childhood and post-reproductive phase, which enables the intergenerational transmission of narratives. This extended timespan gives rise to two rare cognitive capacities:

- The ability to accumulate long-term narrative memory (e.g., through written language, myths, and storytelling);
- The capacity to exert long-term influence on others' behavior and cognition;
- The ability to construct abstract norms based on fictional premises (e.g., legal systems, belief structures, and national identity).

As a result, language becomes more than a communication tool—it evolves into an imprinting device for collective memory, encoding values, boundaries, and belief systems in a stable, transmissible format (Jablonka & Lamb, 2005).

7.4 Interwoven Models of Plot Comprehension and Emotional Language Construction

Taken together, narrative engagement not only enhances linguistic competence but also deepens the internal architecture of emotional models. Characters' decisions, conflicts, and transformations serve as experimental simulations for emotional reasoning, and this simulated experience is eventually embedded back into the expressive structures of language.

According to Barsalou's (2008) Situated Simulation Theory, human understanding of concepts and experiences is not achieved through abstract computation but through multimodal simulation that reactivates prior experiences. Stories and narrative scenarios serve as vessels for this simulation process, enabling individuals to pre-enact emotional and inferential responses in imagined settings.

Likewise, narrative psychology highlights language's core function in constructing and communicating self-experience. As Bruner (1990) argues, stories provide a scaffold for individuals to interpret social norms, select behavioral strategies, and establish identity. This view is consistent with the central claim of this study: that language functions as a tool for managing internal emotional contradictions through narrative structure.

The present study also proposes that this narrative scaffolding explains why humans developed the capacity for fictional cognition—the ability to create imaginary concepts and construct symbolic institutions. Because humans are uniquely able to simulate and transform self–other

dynamics within narrative contexts, they surpass the limitations of concrete experience and develop symbolic structures such as myth, law, and national identity.

In the field of language learning and education, contextualized language input has been shown to be the most effective means of guiding children's understanding of semantics and pragmatics. By engaging with emotionally tense, character-driven plotlines, children are able to safely rehearse linguistic responses to complex social situations, thereby developing their emotional insight, role reasoning, and narrative strategy formation.

In sum, narrative texts can be seen as “low-risk, high-tension simulation laboratories” for language evolution—or what we might call “emotional tension modeling fields.” Within these fields, individuals learn how to translate inner tensions into comprehensible narrative forms, thereby shaping the logic, depth, and flexibility of language itself. Consequently, the density and complexity of narrative exposure during development should be treated as a core predictive factor for linguistic reasoning and narrative capacity (Kidd & Castano, 2013; Mar et al., 2009).

7.5 Summary and Extended Perspective: Language as a Narrative Device for Simulating Self and Society

The driving force behind language evolution stems not only from the need to “say something,” but also from the imperative to “construct and simulate how something is said—and to whom.” As the latest and most culturally influential dimension of language evolution, narrative capacity integrates multiple factors, including memory imprinting, social structure, temporal cognition, and value transmission.

The simulation structure inherent in narrative language not only shapes expressive strategies but also progressively refines the granularity of emotional experience. This dynamic co-evolution between language and emotion forms a reciprocal feedback loop—language refines emotional categories, while emotional tension demands new linguistic constructions (see also Chapter 9 for extended analysis).

Chapter 8 | Exploring Interdisciplinary Application Potential: From Language Education and Psychotherapy to Artificial Intelligence

This chapter explores the practical and interdisciplinary applications of the Emotion-Driven Language Evolution Model across three major domains: language education, clinical psychology, and artificial intelligence. The theoretical foundation laid in previous chapters emphasizes that language emerged not merely from the need to convey information, but from the urgent necessity to process emotional tension and internal contradiction—stimulating physiological structures and, through narrative simulation, enabling the conversion of intent into structured expression. Here, we extend these insights into concrete mechanisms and potential application blueprints.

8.1 Contextual Narrative Density and the Neurodevelopmental Fit Model for Language Reasoning

Language acquisition is not a linear process of syntactic mastery; rather, it involves the progressive development of skills for resolving narrative tension and simulating emotional states. According to the five-level model of emotional complexity (L1–L5) and the triadic generation model ($E_c \times C_n \times N_s$) proposed in this study, the density and complexity of narrative input directly influence a child’s development in linguistic reasoning and social cognition.

The following chart integrates the L1–L5 emotional complexity model (as defined in Section 6.1), aligning emotional levels with linguistic challenges and neural developmental stages:

Age Group	Suggested Text Features	Emotional Tier	Reasoning Challenge	Key Brain Regions	Functional Role	Physiological Milestones
2–4 years	Single-layer plots, basic emotions	L1–L3	Basic cause-and-effect linking	Temporal language areas, early-stage PFC	Vocabulary acquisition, reactive connections	Vocabulary explosion period
5–7 years	Role misinterpretations, emotional conflict	L2–L4	Theory of Mind reasoning (ToM)	Right TPJ, anterior cingulate cortex (ACC)	Perspective-taking, error detection	Theory of Mind formation phase
8+ years	Multi-sequence narratives, value transformation	L4–L5	Role-based motivation, contextual shifts	Medial PFC, dorsolateral PFC (DLPFC)	Inhibition, value hierarchy, strategic planning	Executive function onset

Early intervention in language education, when enriched by emotionally dense and context-rich inputs, offers not only cognitive benefits but also dual-layered developmental protection at both neural and epigenetic levels (see also Section 5.5).

8.1-2 Narrative Simulation Design for Adolescent Language Training: A Cross-Cultural Perspective

Within the global trend of language education, growing attention is being paid to the fact that language is not just a matter of phonological and syntactic instruction—it is also a training ground for narrative reasoning, emotional simulation, and perspective-shifting (Bruner, 1990;

Barsalou, 2008). This study argues that effective language instruction for adolescents must include a mid-level cognitive bridge based on “emotional modeling × narrative simulation,” especially during the critical shift from L2–L3 to L4.

Cross-referencing the five-level emotional model and the $E_c \times C_n \times N_s$ triadic model, we recommend incorporating high-tension narratives involving perspective shifts and value-based conflicts into curricula. These allow students to move beyond surface-level expressions toward complex pragmatic and reasoning structures.

Preliminary observations from comparative educational studies suggest:

In Nordic and Anglophone countries (e.g., Sweden, Canada, the UK), secondary language curricula already systematically integrate writing and speaking modules focused on perspective-taking, contradiction, and inference.

In contrast, many other regions still emphasize vocabulary, rhetorical devices, reading comprehension, and grammar drills, creating a developmental gap in high-level narrative construction and cognitive simulation.

This study therefore recommends that language education shift its emphasis from output format and lexical expansion to exercises that invoke subjective tension, role understanding, and narrative logic, thereby reinforcing the core educational function of language as a tool for simulation-based reasoning.

8.1-3 The Adverse Effects of Fragmented Narratives on Language and Simulative Thought

In today’s linguistic environment, short-form videos, slogan-like posts, and broken dialogues dominate the daily language input of adolescents. These de-narrativized stimuli may accelerate response times and increase information throughput but significantly impair the training of perspective-shifting, emotional contradiction processing, and long-form reasoning.

According to this study’s model, when language input is restricted to L1–L2 emotional levels and lacks temporal scope or multiperspectival narrative challenges, the individual’s developmental potential within the $E_c \times C_n \times N_s$ triad is likely to be suppressed.

Such effects are already observable in subsets of adolescent populations, manifesting as disconnected language expression, poor emotional logic, and narrative impoverishment—a new class of sociopragmatic impairments.

Consequently, both educational and digital media policy should proactively incorporate interventions that offer deep narrative structures and simulation challenges, as a counterbalance to the degenerative trends induced by short-form media.

Contemporary publications such as *The Shallows* (Carr, 2010) and *The Whole-Brain Child* (Siegel & Bryson) have already highlighted how fragmented digital environments undermine deep thinking, narrative patience, and self-construction in youth. These concerns strongly align with the current study's "emotion–reasoning–narrative chain model," positioning simulative language decay as an urgent structural challenge facing modern education and digital culture.

8.2 Emotional Blunting and Linguistic Closure: Narrative Reframing as Intervention for Affective Disorders

Clinical observations indicate that patients with depression or anxiety often exhibit slowed verbal response, internal semantic blockage, emotional vocabulary deficiency, and repetitive thinking loops. These symptoms suggest that individuals may be stuck in L1–L2 emotional levels, unable to transition into the narrative and reasoning stages of L3–L4—particularly in terms of processing emotional contradiction within social contexts.

This study proposes a Structured Language Reframing (SLR) intervention framework, in which narrative restructuring serves as a psycholinguistic tool for mental health support:

- Emotional Vocabulary Expansion: Moving from single emotional states (e.g., "tired," "afraid") to layered phrases (e.g., "guilt in fatigue," "desire within fear").
- Narrative Progression Training: Guiding clients into simulations of "what I want to say but can't" or "what happens if I don't speak."
- Perspective Broadening: Encouraging simulation of others' interpretations of one's silence or statements—constructing social feedback frames.

This process can be supported by visual tools (e.g., character maps, event chains) to strengthen the client's capacity for contextual encoding and may serve as a bridge between narrative therapy and CBT.

8.3 Emotional Language Generation and Narrative Simulation in Artificial Intelligence

Contemporary large language models (LLMs) generate text primarily based on statistical association and semantic embedding but struggle to replicate the deeper simulation processes associated with L3–L4 emotional complexity—namely, "multi-emotional tension × presupposed role strategies × contextual adaptation."

This study suggests that future AI models aiming for higher-level pragmatics and subjective inference must incorporate a dual-extension based on E_c and N_s .

- Proposed mechanisms include: E_c Embedding: Embedding multi-level emotional structures and hierarchical need codes into instruction architecture, allowing models to detect and reframe emotional contradictions (e.g., irony, suppression, tension) via syntax modulation and word order manipulation.

- Narrative Simulation Layer: Implementing modules capable of handling temporal discontinuity, role perspective switching, and misaligned intent repair within output sequences.

- Language Potential Estimation via Formula: Applying the model's formula $L_g = f(E_c \times C_n \times N_s)$ as a metric for guiding output strategies and simulation depth.

If future systems can accurately simulate behaviors such as hesitation and message delay under shame, or subject-switching under conflicting intentions, then AI will have entered an era of genuine pragmatic logic construction.

8.4 Summary: Toward Simulated Language Modeling in Education, Psychology, and AI

This chapter began with three typical application scenarios and demonstrated that the Emotion-Driven Language Evolution Model possesses remarkable adaptability and theoretical translatability. Its underlying logic can be summarized as follows:

- Language generation is not reflexive, but a dynamic structure for tension regulation and simulation pressure;

- Narrative is not merely a form of content organization, but a force field for psychological imprinting and social strategy formation;

- The linkage between emotional modeling and linguistic simulation is quantifiable, designable, and operational.

This model has the potential to catalyze integrative research across five domains: language science, educational intervention, clinical psychology, language-based AI systems, and civilizational narrative evolution.

Chapter 9 | Conclusion, Academic Contributions, and Theoretical Positioning

9.1 Core Contributions and Theoretical Positioning of This Study

This study proposes emotional complexity as the central driving force in the evolution of human language, systematically constructing two integrated models: the Emotion–Intention–Reasoning Chain and the triadic model of $E_c \times C_n \times N_s$. Through multi-layered empirical frameworks—including cross-species comparison, neural mechanisms, and behavioral indicators—this hypothesis offers a verifiable and applicable theoretical framework for language origins.

The primary academic innovations of this study include:

- **Theoretical Compensation:** For the first time, it frames emotional complexity and internal tension as the principal axis of pressure for language generation. It addresses the blind spots in the Tool Hypothesis, Social Bonding Hypothesis, and Cognitive Niche Hypothesis, especially regarding emotion, narrative, and subjective reasoning.
- **Operational Models:** It establishes a five-level emotional complexity framework (L1–L5) and the $E_c \times C_n \times N_s$ interaction model, both with strong potential for quantification and cross-species application—providing specific metrics for research in evolutionary linguistics and cognitive neuroscience.
- **Interdisciplinary Integration:** The study effectively integrates insights from linguistics, neuroscience, psychology, evolutionary biology, AI, and education, offering a common operational language across traditionally fragmented disciplines.
- **Application Prospects:** It demonstrates that the model may extend into domains such as special education, clinical treatment, and AI narrative modules—thus breaking through the purely descriptive limits of traditional language evolution theories.

9.2 Comparative Theories and the Position of This Hypothesis

To clarify the distinctions and complementarities between the current study and existing mainstream theories of language origins, the table below summarizes the driving forces, mechanisms, strengths, limitations, and contributions of each model:

Theory	Primary Driver	Mechanism	Strengths	Limitations	Contribution of This Hypothesis
Tool-Use Hypothesis	Motor planning, hand coordination	Sequential logic, syntax	Action–syntax overlap; Broca’s area correlation	Lacks emotional origin and symbolic inference	Adds symbolic pressure driven by emotional differentiation
Social Bonding Hypothesis	Group cohesion	Vocal grooming, lexical repetition	Explains language’s social function	Lacks recursive logic and narrative tension	Bridges social bonding with emotional contradiction modeling
Cognitive Niche Theory	Environmental modeling	Symbolic abstraction, prediction	Strong on reasoning and abstract modeling	Neglects emotion and narrative as dynamic sources	Adds <i>narrative simulation</i> and <i>emotion-based intent inference</i>
Mirror Neuron Hypothesis	Action imitation, intention modeling	Mirror neuron simulation	Explains motor–language transition	Cannot explain semantic recursion and emotional logic	Emphasizes emotional modeling as the critical threshold
Musical Origins Theory	Vocal modulation, rhythmic prosody	Gradual transition to speech	Accounts for tone and emotional signaling origins	Lacks syntax and semantic scaffolding	Clarifies rhythm-to-language transition via emotional contradiction
Symbolic Species Theory	Cultural transmission	Symbolic abstraction and iteration	Explains co-evolution of language and brain	Lacks biological and neuroplasticity evidence	Integrates emotion, society, and neural evolution into one model

As discussed in Chapter 4, this theory was initially conceived as a complementary framework to mainstream hypotheses. However, by the time we arrive at the operational constructs in Chapters 5–7—such as emotional tiering, linguistic pressure equations, and narrative simulation capacity—the potential of this model for integrative synthesis becomes increasingly apparent.

9.3 Interdisciplinary Contributions and Future Research Trajectories

Academic Impact:

- This hypothesis offers a novel pathway in language origin research by positioning emotional modeling as the evolutionary watershed—bridging gaps in neurolinguistics, developmental psychology, and cognitive science.
- It provides a framework for comparing emotional behavior indicators across species (mammals, birds, social insects), enabling empirical validation through cross-domain models.
- For AI and narrative engine design, it offers a principled structure for embedding emotional modeling and tension regulation into next-generation language systems.

Applied Research Implications:

- In language education, special education, and clinical therapy, the model offers new diagnostic and intervention metrics through emotional tiering and narrative simulation assessment.
- In AI and digital storytelling, it provides a design framework for simulating “emotion × narrative × reasoning” in generative language systems, enabling richer, more human-like outputs.
- In brain–machine interface and human–AI interaction, it lays a theoretical foundation for multimodal emotional-language encoding.

Future Directions and Open Questions:

- Cross-Species Empirical Design: Testing the $E_c \times C_n \times N_s$ model across mammals, birds, and AI systems to evaluate its predictive power and scalability.
- Neural Plasticity and Epigenetics: Combining fMRI and gene expression analysis to explore how emotional tension × language generation leaves imprints at the molecular and neurological levels.
- Narrative Intervention Studies: Examining how exposure to rich contextual narratives affects linguistic and inferential development in children, adolescents, and language-impaired populations.
- AI System Prototyping: Embedding the triadic model into language generation architectures to test improvements in narrative coherence, emotional logic, and subjective inference.
- Civilizational Narrative Evolution: Analyzing myth, religion, and national storytelling traditions for patterns of co-evolution between emotional complexity and linguistic structure.

9.4 Closing Remarks

Although many great thinkers—such as Jablonka & Lamb (2006), Barrett (2017), and Deacon (1997)—have touched on similar ideas, this study attempts to consolidate these fragmented perspectives into a unified explanatory framework. It aspires to provide a foundation for cross-species, cross-disciplinary inquiry into the origins of language and cognition. As emotional tension and narrative capacity gain increasing relevance in AI, human–machine collaboration, and cultural evolution, the Emotion-Driven Hypothesis for Language Evolution may well become a foundational reference for future studies in language and intelligence.

Originally built upon the concept of “emotional complexity” as a key pressure source, and developed through the $E_c \times C_n \times N_s$ triadic model, this theory ultimately reveals a deeper insight: language and emotion did not evolve in a unidirectional causal chain but rather in a dynamic, reciprocal feedback loop.

Each increase in emotional granularity demanded more precise symbolic articulation; each narrative advancement in language restructured and refined emotional awareness in turn. Over time, this recursive interaction forged a co-evolutionary system of language and emotion—a process not merely biological or cultural, but one that involved the iterative upgrading of social, neural, and narrative systems in concert.

This model—tentatively called the Co-evolutionary Model of Language and Emotional Complexity—warrants further refinement by future researchers, including those working in neuroethology, comparative linguistics, AI emotion modeling, and civilizational semiotics. Its ultimate value lies in providing new observational tools and theoretical architecture for understanding how intelligence, emotion, and culture emerge from the intertwined pathways of language.

As this theory enters broader academic discourse, it is likely to inspire wide-scale empirical testing, model iteration, and interdisciplinary exploration. For well-equipped research teams, this presents an open challenge—and for this paper, a potential point of origin.

Chapter 10 | References

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Chapter 11 | Postscript: Knowledge Construction in the Era of AI Collaboration

This paper was independently conceived and executed by a guitar teacher with no formal academic training in linguistics. With the support of AI, and drawing on a sensitivity honed through music education and a habit of interdisciplinary exploration, the author has woven together insights from language, emotion, neuroscience, and education to produce a comprehensive academic work. This project demonstrates that theoretical innovation need not be confined to traditional academic pathways. In an era empowered by AI and open-access knowledge, anyone with a passion for inquiry has the opportunity to participate in scholarly discourse.

It all began with a simple intuition: Could language have originated from the increasing complexity of emotional modeling? As a lifelong learner endlessly curious about knowledge itself, I suspected that perhaps humanity had already uncovered fragments of this trajectory. My role was not to invent a destination, but rather to bring these fragments together—equipped only with a computer, a few bottles of tea, a stable internet connection, and the companionship of an

AI assistant. The process was one of hypothesis generation, iterative deduction, and continual revision.

In the early drafting stage, I posed my core hypothesis and central questions to an AI (ChatGPT), which then generated a preliminary table of contents, chapter outlines, and an introductory framework based on my inputs. I proceeded to write the manuscript chapter by chapter—revisiting, rethinking, refining—continually adding new perspectives, proposing counterarguments, and strengthening the reasoning. Sometimes I asked the AI for suggestions, sometimes I had it review the logic of my arguments, and sometimes we simply talked. There were also the dozens—perhaps hundreds—of times I asked it to re-evaluate the entire text for coherence, scientific rigor, and internal consistency. This practice of “extending the cognitive bandwidth of the human brain through AI” significantly enhanced both the efficiency and logical depth of my writing. The result was a self-critical, evolving paper that drew simultaneously from linguistics, neuroscience, cognitive psychology, artificial intelligence, and education.

There were no towering library shelves, no long-standing academic teams. Only the infinite stream of information in the digital age, a learner willing to challenge disciplinary boundaries, and an AI system acting as a cognitive mirror. Perhaps this is a glimpse of how knowledge will be produced in the future: with a sense of inquiry, a willingness to experiment, and the ability to leverage the tools of our time, anyone can create a work of disruptive insight in just a matter of weeks.

I am grateful for every bottle of tea and every line of code that made this possible. This, too, is a miracle of creativity—one that belongs uniquely to our era.

Again, Note to readers:

Note to readers:

As a non-native English speaker and independent researcher from Taiwan, I sincerely welcome any feedback, corrections, or suggestions.

If you find the hypothesis presented in this paper of interest and would be willing to assist me in submitting it to arXiv, my endorsement code is **CLXFOS**.

For any further inquiries, interest, or discussion, please feel free to contact me at:

jix1001@yahoo.com.tw

Your time, insight, and generosity are deeply appreciated.:

— **Chun-Hao Chan**